COMPONENTS

A smartphone chipset with on-device AI

MediaTek has launched the Dimensity 9600 Pro, a 2nm flagship smartphone chipset built for on-device artificial intelligence (AI). It integrates dual neural processing units (NPUs) and the Dimensity Agentic AI Engine to understand user context and coordinate tasks across apps. The G2-Ultra NX GPU handles gaming, while the Imagiq 1290 processor enhances photo and video. The OPPO Find X10 Pro Max will be the first Indian device to incorporate it.



MediaTek
https://www.mediatek.com

A low-power SoC for small connected devices

Nordic Semiconductor has introduced the nRF54LC10A, a low-power multi-protocol SoC for compact connected devices. It supports Bluetooth Low Energy (BLE), Bluetooth Channel Sounding, Thread, Zigbee, and Matter on a single chip, saving board space and cost. Channel Sounding enables distance measurement and positioning, while Thread radio coprocessor support suits home hubs and border routers. Applications include asset trackers, smart-home accessories and connected sensors.



Nordic Semiconductor
https://www.nordicsemi.com

A high-speed ADC to catch data centre faults

Texas Instructions has introduced the ADC34DSO75, a high-speed application delivery controller (ADC) that detects brief power-supply disturbances in AI data centres. Fast sampling and integrated digital signal processing capture transient voltage faults that conventional monitoring may miss. It enables continuous, remote monitoring of low-voltage, high-current rails without on-site oscilloscopes, improving power-delivery reliability and reducing downtime in AI servers and accelerator systems.



Texas Instruments
https://www.ti.com

A new 65V digital power monitor

Microchip Technology has introduced the PAC1761 and PAC1861 digital power monitors for 48V systems in AI data centres, automotive, servers and industrial equipment. They measure up to 65V and offer transient spike protection and accumulated-energy tracking. Programmable alerts flag voltage, current, and power excursions, while step-limit detection spots sudden load changes. Pin-compatible options simplify design flexibility and upgrades.



Microchip Technology
www.microchip.com

An MCU supporting dual displays for vehicles

Infineon Technologies has introduced the TRAVEO T2G CYT4EN MCU for automotive instrument clusters and cockpit displays, including two-wheelers. It supports dual Full HD displays and 3D graphics without a more complex system-on-chip (SoC), with external LPDDR4 memory and startup in under 150ms. Integrated functional safety and cybersecurity features meet automotive requirements, while reduced component count and cooling needs contribute to lower cost.



Infineon Technologies
https://www.infineon.com

An AI processor for continuous vehicle surveillance

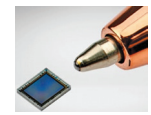
OMNIVISION has introduced the OAX7700, an ultra-low-power AI companion processor for continuous vehicle surveillance without keeping the main computing platform awake. Its integrated NPU analyses camera feeds locally to detect people, objects, and security events, retaining pre-event video for the Electronic Control Unit (ECU) to review. Customised AI algorithms and over-the-air updates support flexible parking surveillance and software-defined vehicles while cutting power consumption.



OMNIVISION
https://www.ovt.com

A compact IR sensor for in-cabin monitoring

STMicroelectronics has introduced the ST SafeSense VD56GA, a compact automotive image sensor for driver and occupant monitoring. Its DeepNIR backside-illuminated pixel architecture improves infrared sharpness and sensitivity, including for cameras placed behind displays. Integrated image processing reduces reliance on external resources, and RAW and YUV outputs suit different vehicle architectures, supporting cost-effective in-cabin sensing.



STMicroelectronics
https://www.st.com

A new tiny reprogrammable mixed-signal IC

Renesas Electronics has introduced the SLG46801 GreenPAK, a compact configurable mixed-signal IC that adds reprogrammable logic to space-constrained designs. Engineers can customise digital logic functions without redesigning the whole

